Visualization and Manuscript Injection

Figure Generation I

Figures are rendered headlessly (matplotlib Agg backend) and deterministically from the analysis artifacts: subfield distributions, the publication growth curve, the citation network, topic-term bars, a term cloud, and embedding projections. All figures use a colourblind-safe palette (Wong 2011, 8 colours) with high-contrast labels at ≥ 16pt. This run produced 18 figures at 300 DPI. The full figure set includes:

- **Field overview**: field summary and subfield distribution ((Figure field summary; Figure subfield distribution))
- **Temporal**: growth curve and subfield timeline ((Figure growth curve; Figure subfield timeline))
- **Descriptive**: citation distribution, top venues, and author productivity ((Figure citation distribution; Figure top venues; Figure author productivity))
- **Citation network**: network layout and degree distribution ((Figure citation network; Figure degree distribution))

Figure Generation II

- **Hypothesis**: evidence dashboard ((Figure hypothesis dashboard))
- **Text analytics**: word cloud, topic-term bars, PCA embeddings, term heatmap, dendrogram, and co-occurrence matrix ((Figure word cloud; Figure topic term bars; Figure pca embeddings; Figure term heatmap; Figure dendrogram; Figure cooccurrence matrix))
- **Entities**: named-entity bar chart and similar-document pairs ((Figure entity bar chart; Figure similarity heatmap))

Each figure is registered in `figure_registry.json` with its label, caption, filename, and generating stage, binding the visual output to the analysis artifacts of the exact pipeline run.

Variable Injection I

The manuscript itself is generated, not hand-maintained. A variable computation step reads the configuration and the pipeline outputs and emits a flat table of named values; an injection step substitutes each named placeholder in these Markdown sections with its computed value before rendering. Because the substitution is total, an unresolved placeholder is a hard error rather than a silent gap. Every number in the rendered document is guaranteed to trace to a committed artifact. Re-running the pipeline after a configuration change re-computes the values and re-targets the prose automatically.

The injection system computes variables from the configuration and generated artifacts, including:

1. `manuscript/config.yaml`: search term, engine roster, subfield taxonomy, hypotheses
2. `corpus.jsonl`: corpus size

Variable Injection II

3. `temporal_analysis.json`: year range, CAGR, peak year, doubling time
4. `citation_network.json`: edges, nodes, density, communities, PageRank, hubs
5. `subfield_classification.json`: per-bucket counts and percentages
6. `assertion_summary.json`: assertion counts and directions
7. `hypothesis_scores.json`: per-hypothesis evidence scores